

# LAB 1

DevOps Foundations & Continuous Integration

## Task 4 - Install and Configure Git and Jenkins

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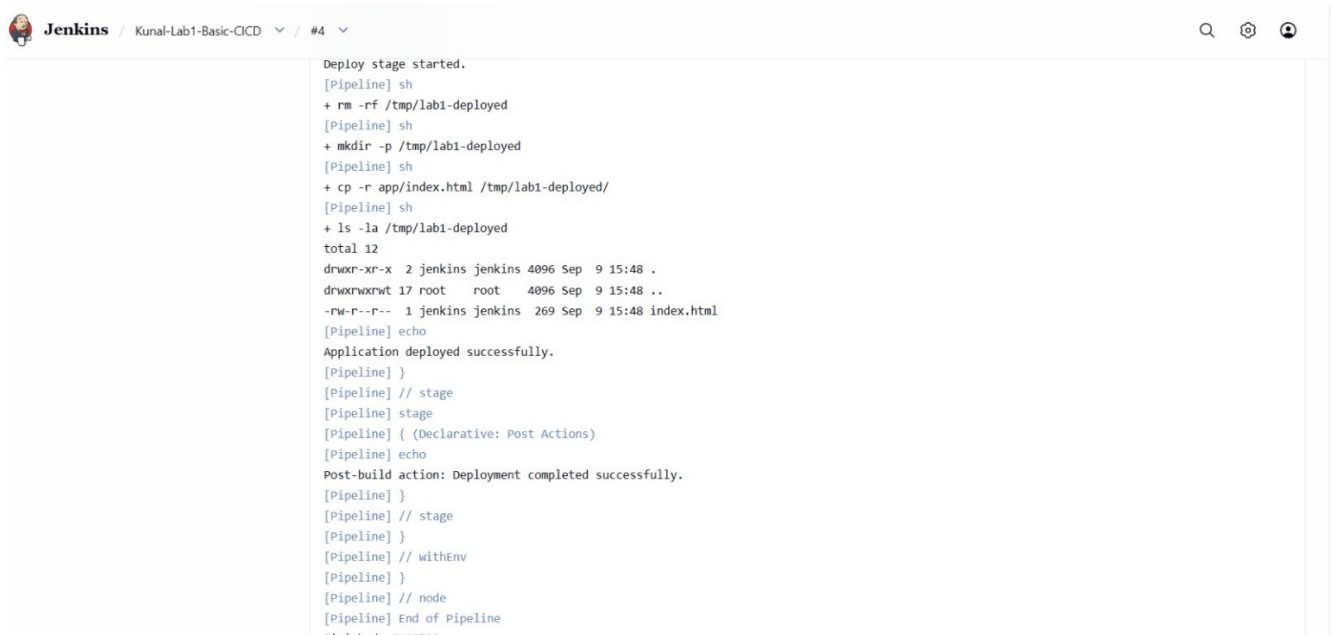
### Objective

Install and configure Git on Ubuntu Linux, verify the installation, configure the global Git identity, install Java required by Jenkins, install Jenkins, and verify that the Jenkins service is enabled and running.

**Environment:** Ubuntu 24.04.4 LTS (WSL2)   **Git:** 2.43.0   **Jenkins:** 2.568.3   **Java:** OpenJDK 21

# 1. Ubuntu Linux Environment and Student Verification

Ubuntu 24.04.4 LTS was verified using **lsb\_release -a**. The terminal also records the student label, current Linux user, and date. This establishes the working environment used for the lab.



```
Jenkins / Kunal-Lab1-Basic-CICD / #4
Deploy stage started.
[Pipeline] sh
+ rm -rf /tmp/lab1-deployed
[Pipeline] sh
+ mkdir -p /tmp/lab1-deployed
[Pipeline] sh
+ cp -r app/index.html /tmp/lab1-deployed/
[Pipeline] sh
+ ls -la /tmp/lab1-deployed
total 12
drwxr-xr-x 2 jenkins jenkins 4096 Sep  9 15:48 .
drwxrwxrwt 17 root    root    4096 Sep  9 15:48 ..
-rw-r--r-- 1 jenkins jenkins 269 Sep  9 15:48 index.html
[Pipeline] echo
Application deployed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Post-build action: Deployment completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Figure 1. Ubuntu 24.04.4 LTS, student label, username, and date verification.

## 2. Git Installation and Version Verification

Git was installed and its version was verified successfully. The terminal output reports **git version 2.43.0**, confirming that Git is available for version-control operations.

```
kunal_prajapati@LAPTOP-45509TK7:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
kunal_prajapati@LAPTOP-45509TK7:~$ echo "Student Name: Kunal"
Student Name: Kunal
kunal_prajapati@LAPTOP-45509TK7:~$ whoami
kunal_prajapati
kunal_prajapati@LAPTOP-45509TK7:~$ date
Wed Sep  9 12:33:13 UTC 2026
kunal_prajapati@LAPTOP-45509TK7:~$
```

Figure 2. Git version verification showing Git 2.43.0.

### 3. Global Git Configuration

The global Git configuration was verified using **git config --global --list**. The configured global identity is shown as **user.name=Kunal** and the configured email address is displayed in the terminal evidence.

```
kunal_prajapati@LAPTOP-45509TK7:~$ echo "Student Name: Kunal"
Student Name: Kunal
kunal_prajapati@LAPTOP-45509TK7:~$ git --version
git version 2.43.0
kunal_prajapati@LAPTOP-45509TK7:~$ |
```

Figure 3. Global Git user name and email configuration.

## 4. Java Runtime Installation

OpenJDK 21 was installed and verified using **java --version**. The terminal evidence shows OpenJDK version 21.0.12 on Ubuntu 24.04, providing the Java runtime required for the Jenkins installation used in this lab.

```
kunal_prajapati@LAPTOP-45509TK7:~$ git config --global --list
user.name=Kunal
user.email=kunalprajapati14042005@gmail.com
kunal_prajapati@LAPTOP-45509TK7:~$ |
```

Figure 4. Java 21 runtime verification.

## 5. Jenkins Installation and Version Verification

Jenkins was installed and its version was checked using **jenkins --version**. The evidence reports Jenkins version **2.568.3**.

```
kunal_prajapati@LAPTOP-45509TK7:~$ echo "Student Name: Kunal"
Student Name: Kunal
kunal_prajapati@LAPTOP-45509TK7:~$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-24.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-24.04-Ubuntu, mixed mode, sharing)
kunal_prajapati@LAPTOP-45509TK7:~$ |
```

Figure 5. Jenkins version verification showing Jenkins 2.568.3.

## 6. Jenkins Service Enabled and Running

The Jenkins service was enabled and started using systemd. The status output shows **enabled** and **Active: active (running)**, confirming that the Jenkins Continuous Integration Server is running correctly.

```
kunal_prajapati@LAPTOP-45509TK7:~$ echo "Student Name: Kunal"
Student Name: Kunal
kunal_prajapati@LAPTOP-45509TK7:~$ jenkins --version
2.568.3
kunal_prajapati@LAPTOP-45509TK7:~$ |
```

Figure 6. Jenkins systemd service enabled and active.

## 7. Jenkins Dashboard Verification

The Jenkins web interface was opened at **http://localhost:8080**. The Welcome to Jenkins dashboard confirms successful access to the installed Jenkins server.

```
kunal_prajapati@LAPTOP-45509TK7:~$ sudo systemctl status jenkins --no-pager
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-09 12:46:50 UTC; 1min 50s ago
     Main PID: 2130 (java)
       Tasks: 50 (limit: 8276)
      Memory: 1.0G ()
     CGroup: /system.slice/jenkins.service
             └─2130 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkin...

Sep 09 12:46:46 LAPTOP-45509TK7 jenkins[2130]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initial...assword
Sep 09 12:46:46 LAPTOP-45509TK7 jenkins[2130]: [LF]>
Sep 09 12:46:46 LAPTOP-45509TK7 jenkins[2130]: [LF]> *****
Sep 09 12:46:46 LAPTOP-45509TK7 jenkins[2130]: [LF]> *****
Sep 09 12:46:46 LAPTOP-45509TK7 jenkins[2130]: [LF]> *****
Sep 09 12:46:50 LAPTOP-45509TK7 jenkins[2130]: 2026-09-09 12:46:50.846+0000 [id=61] INFO jenkins.I...ization
Sep 09 12:46:50 LAPTOP-45509TK7 jenkins[2130]: 2026-09-09 12:46:50.869+0000 [id=38] INFO hudson.li...running
Sep 09 12:46:50 LAPTOP-45509TK7 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Sep 09 12:46:52 LAPTOP-45509TK7 jenkins[2130]: 2026-09-09 12:46:52.948+0000 [id=77] INFO h.m.Downl...staller
Sep 09 12:46:52 LAPTOP-45509TK7 jenkins[2130]: 2026-09-09 12:46:52.950+0000 [id=77] INFO hudson.ut...empt #1
Hint: Some lines were ellipsized, use -l to show in full.
kunal_prajapati@LAPTOP-45509TK7:~$
```

Figure 7. Jenkins dashboard accessed successfully at localhost:8080.

## 8. Commands and Observations

### Key commands executed

```
lsb_release -a
echo "Student Name: Kunal"
whoami
date
sudo apt update
sudo apt install git -y
git --version
git config --global user.name "Kunal"
git config --global user.email "kunalprajapati14042005@gmail.com"
git config --global --list
sudo apt install fontconfig openjdk-21-jre -y
java --version
sudo apt install jenkins -y
jenkins --version
sudo systemctl enable --now jenkins
sudo systemctl status jenkins --no-pager
```

### Observations

1. Ubuntu Linux 24.04.4 LTS was used as the lab environment.
2. Git was installed successfully and its version was verified.
3. Global Git user name and email were configured successfully.
4. OpenJDK 21 was installed and verified for Jenkins operation.
5. Jenkins 2.568.3 was installed successfully.
6. The Jenkins systemd service is enabled and active (running).
7. The Jenkins dashboard is accessible through localhost:8080.

## Conclusion

Task 4 was completed successfully. Git and Jenkins were installed on Ubuntu Linux, the Git global identity was configured, Java 21 was verified, Jenkins was enabled and running as a system service, and the Jenkins dashboard was successfully accessed.